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Predict, Prevent, Decide: AI-Enabled Root Cause Analysis and Operational Intelligence for Offshore Performance

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Abstract

Unplanned shutdowns remain a persistent challenge in offshore oil and gas operations—causing production disruption, safety risk, and elevated carbon emissions. These events range from minor disruptions (2–4 hours, ~$40,000/event, ~1.5 MT CO_2e flared) to major cascading failures (8–24 hours, $300,000–$500,000/event, 10–15 MT CO_2e flared). Even isolated shutdowns can erode profitability and compromise ESG targets.

Despite the presence of advanced data sources—SCADA, DCS, ETAP simulations, vibration logs, SAP-PM systems, and vendor diagnostics—most platforms struggle to translate this data into predictive and economically prioritized maintenance action. Signals remain siloed, fault chains go unrecognized, and decision-making often defaults to time-based or alarm-triggered responses. The result is a maintenance strategy that is reactive, inefficient, and misaligned with real-world asset risk.

This paper presents a unified AI-enabled framework for root cause analysis and intelligent maintenance decision-making, termed the Maintenance Causality Loop (MCL). The MCL integrates early fault detection, risk quantification, and task scheduling into a closed-loop system designed to shift offshore maintenance from reactive to proactive.

At its core, the MCL consists of two layers:

- An AI-RCA engine that uses the 6M deviation model (Man, Machine, Method, Material, Measurement, Mother Nature), powered by Dynamic Bayesian Networks, Structural Equation Modelling, and Graph Neural Networks to detect and trace fault propagation
- A decision support layer incorporating a Causal Memory Model, Neglect-Risk Scoring, Return-on-Prevention (RoP) economics, and a Reinforcement Learning–based Smart PM Scheduler

The framework was validated using synthetic datasets representing compressor units, HVAC systems, and emergency diesel generators. Simulations demonstrated a 60% reduction in projected shutdown frequency, >90% PM compliance, and a 1.4× increase in cost-efficiency of preventive maintenance.

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MCL is proposed as a next-generation offshore maintenance strategy—one that unifies diagnostic intelligence with operational execution, extends equipment life, and aligns maintenance actions with both uptime and ESG imperatives.

Introduction

Offshore oil and gas platforms operate under increasingly complex and unforgiving conditions—tight production targets, aging infrastructure, constrained access windows, and growing pressure to meet ESG expectations. Against this backdrop, unplanned shutdowns have become not only cost-intensive but strategically damaging. Each failure event results in lost output, elevated flaring, safety risks, and reputational exposure—making reliability a boardroom priority.

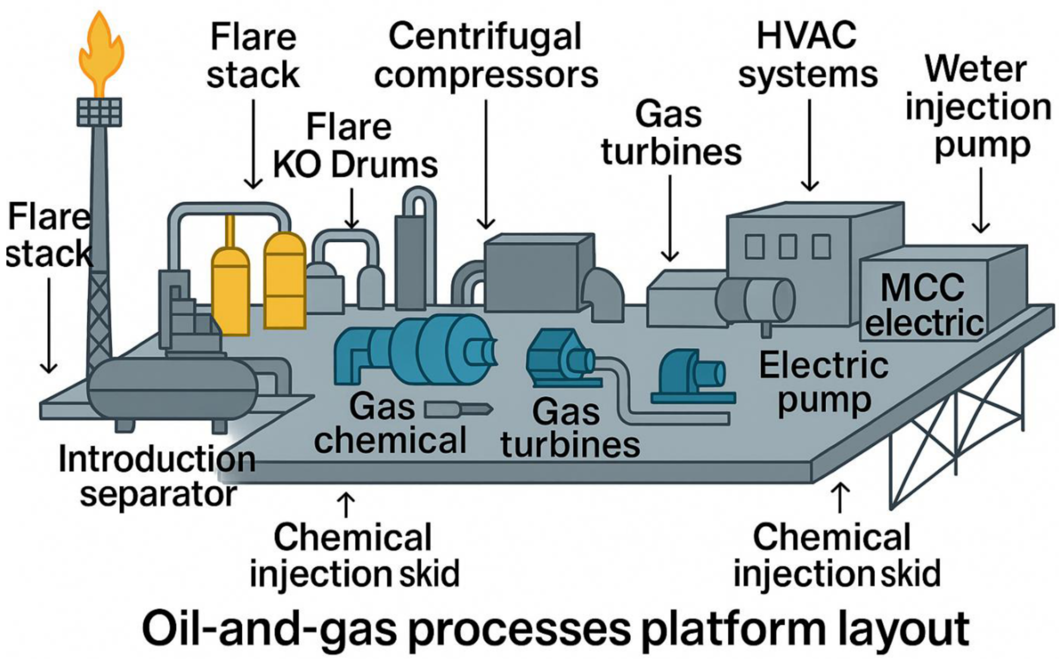

Figure 1—Oil and Gas Process platform layout

Paradoxically, these shutdowns occur despite the availability of vast operational data. Today's offshore assets are deeply instrumented—with SCADA and DCS systems, vibration monitoring, ETAP simulations, SAP-based maintenance logs, and OEM diagnostics. Yet, across most platforms, this data remains underutilized. Critical decisions are still governed by calendar-based maintenance, reactive firefighting, or alarm-based triggers, none of which account for evolving degradation or interdependent fault chains.

Failures rarely result from a single cause. Instead, they evolve gradually, through weak signals that are overlooked in isolation: a skipped preventive task, a drifting sensor, a transient voltage fluctuation, or a procedural override. Existing systems do not correlate these patterns, nor do they prioritize interventions based on emerging risk or economic return.

Several structural challenges contribute to this breakdown:

- Siloed data across SCADA, SAP, and vendor systems impedes unified diagnosis.
- Threshold-based alerts miss early causal patterns.
- Static PM schedules overlook condition-specific urgency.

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- No economic framework exists to weigh risk-reduction value vs task cost.
- Decision-making remains subjective, lacking predictive and prescriptive guidance.

What is missing is not more data—but a decision intelligence layer that can unify, interpret, and act on existing signals to prevent failures before they escalate.

This paper introduces the Maintenance Causality Loop (MCL)—a closed-loop, AI-powered framework that integrates:

- Root Cause Analysis (AI-RCA) using the 6M deviation model and tools like Dynamic Bayesian Networks (DBNs), Structural Equation Modelling (SEM), and Graph Neural Networks (GNNs)
- A **decision support system** incorporating Causal Memory Models, Neglect-Risk Scoring, Return-on-Prevention (RoP) economics, and reinforcement learning-based scheduling.

Together, these layers enable a shift from reactive maintenance to **proactive, risk-aligned, and resource-efficient intervention**, reducing shutdown frequency, extending asset life, and aligning reliability efforts with ESG outcomes.

Statement of Theory and Definitions

This section defines the theoretical structure of the proposed Maintenance Causality Loop (MCL) framework, which integrates root cause diagnostics and preventive maintenance optimization into a closed-loop, AI-augmented system. The MCL comprises two interdependent layers:

- **AI-Enabled Root Cause Analysis (AI-RCA)** — a multi-model engine that ingests high-frequency operational data, classifies deviations, and identifies degradation trajectories and latent fault propagation.

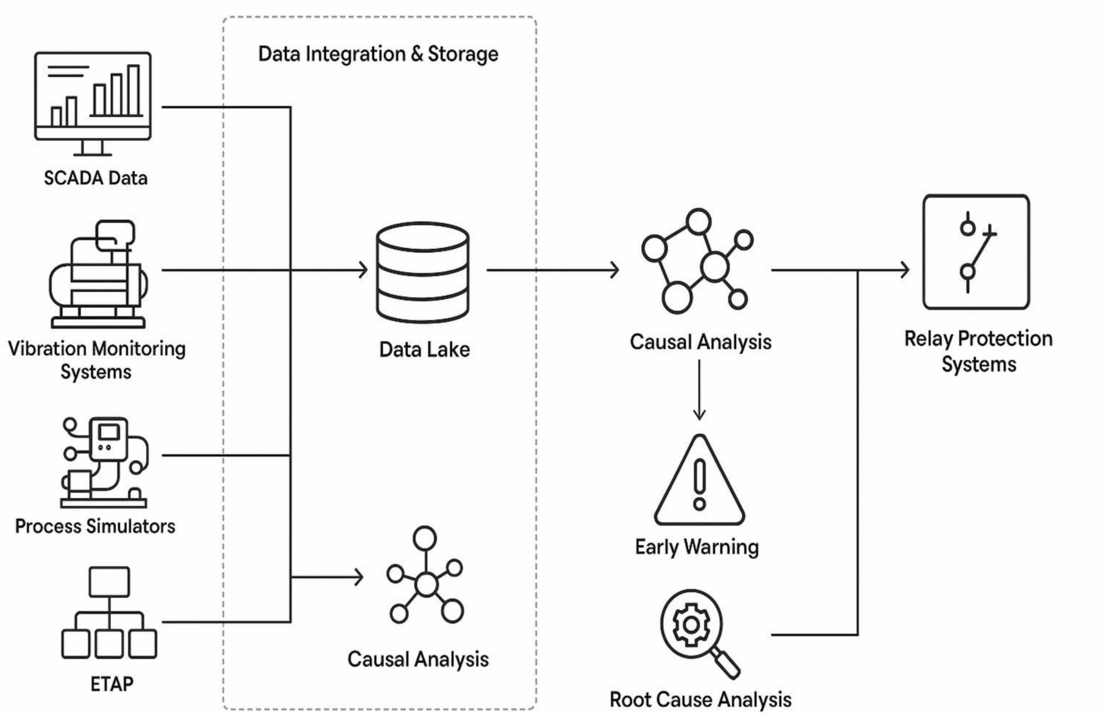

Figure 2—AI-Enabled Root Cause Analysis (AI-RCA)

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- **Decision Intelligence Layer (MCL Engine)** — a prioritization and scheduling module that converts diagnostic output into preventive actions, guided by causal memory, risk metrics, and reinforcement learning optimization.

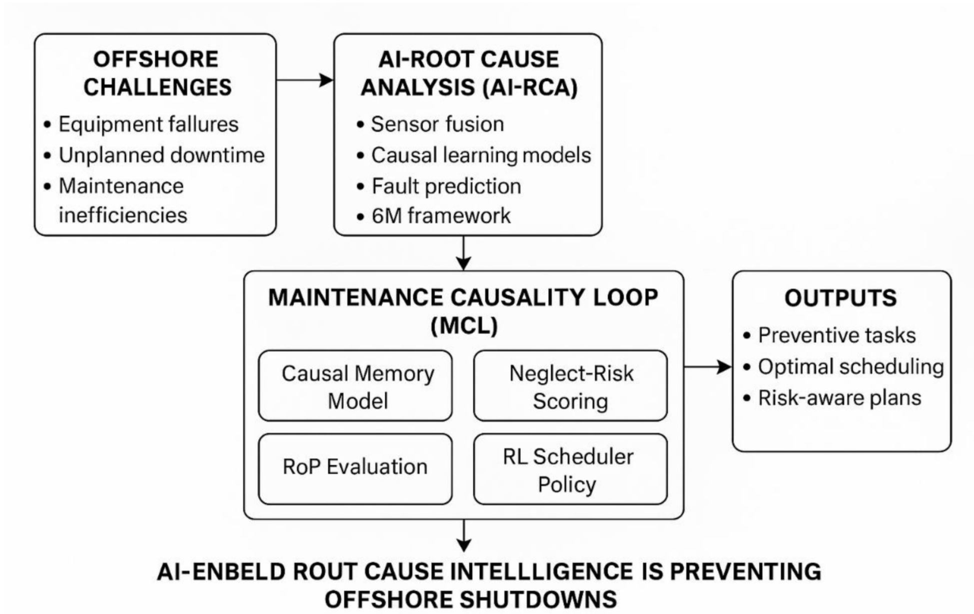

Figure 3—Decision Intelligence Layer (MCL Engine)

Together, these layers transition offshore maintenance from alarm-driven reactivity to data-aligned, risk-prioritized execution.

AI-Enabled Root Cause Analysis (AI-RCA)

The AI-RCA layer serves as the diagnostic core of the framework. It processes multi-source deviation signals—ranging from SCADA, DCS, and vibration logs to SAP-PM records and vendor metadata—and transforms them into structured root cause intelligence using a layered learning pipeline. The analysis is governed by the 6M Framework, which enables categorical attribution of fault signals based on their underlying origin.

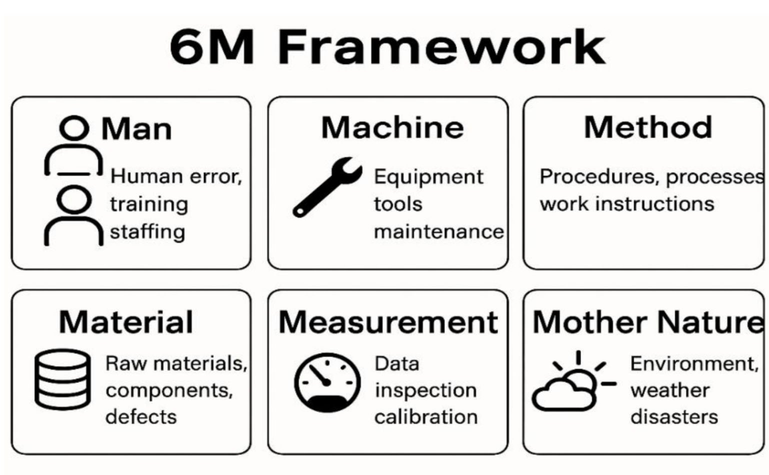

Figure 4—6M Framework

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This structure governs feature extraction and model layering, ensuring each deviation is semantically rooted in its physical or procedural context.

AI-RCA Workflow Architecture

The AI-RCA system transforms offshore operational data into actionable root cause intelligence through a five-stage architecture. Each stage leverages mathematically grounded models to extract, forecast, and explain system deviations. Together, these layers form a scalable diagnostic engine embedded within the 6M framework.

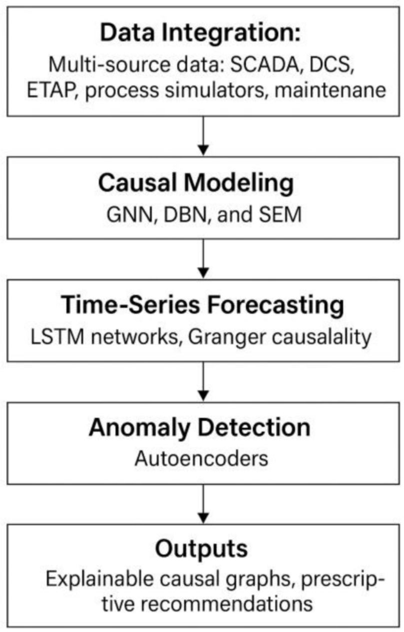

Figure 5—AI-RCA Workflow Architecture

Data Integration

Offshore systems generate fragmented data from SCADA, DCS, ETAP simulations, and maintenance logs. These streams are consolidated into a unified, time-synchronized feature vector:

$$x_t = [x_t{}^{\wedge SCADA}, x_t{}^{\wedge DCS}, x_t{}^{\wedge ETAP}, x_t{}^{\wedge Maint}, \ldots] \qquad \text{Eq-1}$$

This multidimensional vector captures both live telemetry and contextual metadata (equipment class, criticality, vendor history), forming the input foundation for all downstream modeling.

Causal Modeling

This layer identifies the underlying causes behind abnormal patterns by modeling physical dependencies, statistical associations, and latent degradation dynamics.

- Dynamic Bayesian Networks (DBNs) estimate the hidden health state H_t over time, reflecting internal wear:

$$H_{t+1} = \gamma \cdot H_t + (1-\gamma) \cdot g(x_t) \qquad \text{Eq-2}$$

  where γ controls how much weight is given to past wear versus new sensor input.
- Structural Equation Modeling (SEM) infers relationships between latent drivers (e.g., operator error, sub-spec material) and observed faults:

$$Y = \Lambda \cdot \eta + \varepsilon \qquad \text{Eq-3}$$

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This allows the model to handle causality even when root (independent) variables aren't directly measured.

- Graph Neural Networks (GNNs) capture how faults propagate across interconnected equipment or process loops:

$$h_i^{(l+1)} = \sigma\left(\sum_{j \in N(i)} W^{(l)} h_j^{(l)} + b^{(l)}\right) \quad \text{Eq-4}$$

This enables the AI to mirror actual operational hierarchies and failure propagation paths.

Time-Series Forecasting

Once causal chains are detected, the model forecasts signal evolution to pre-empt threshold breaches.

- LSTM (Long Short-Term Memory) networks learn temporal dependencies to predict the future trajectory of variables like temperature, vibration, and flow rate:

$$\hat{x}_{t+\tau} = \mathrm{LSTM}(x_t, x_{t-1}, \ldots, x_{t-n}) \quad \text{Eq-5}$$

  - Granger Causality evaluates whether historical data from one signal predicts another:

$$x_j(t) = \alpha_0 + \sum \alpha_i x_j(t-i) + \sum \beta_i x_k(t-i) + \varepsilon_t \quad \text{Eq-6}$$

This quantifies directional influence between variables (e.g., compressor discharge pressure → seal failure risk).

Anomaly Detection

To catch emerging failures not yet flagged by alarms, the AI uses autoencoder neural networks trained on healthy signal profiles. At each time step:

$$L\_AE = \| x_t - \hat{x}_t \|^2 \quad \text{Eq-7}$$

Where $\hat{x}_t$ is the reconstructed signal. A rising reconstruction error implies deviation from normal operating behavior — a precursor to many shutdowns.

Output Layer – Causal Graphs & Prescriptions

The final output is human-actionable intelligence:

- Causal Graphs visualize the propagation of faults (e.g., from a valve delay to compressor overload).
- Prescriptive Alerts suggest interventions prioritized by risk and effort (e.g., adjust setpoint vs. replace seal).
- Failure Probabilities such as:

$$P(F_{t+}T) = \sigma(\beta_0 + \beta_1 H_t + \beta_2 N_t) \quad \text{Eq-8}$$

These outputs enhance decision confidence and bridge the gap between AI insights and real-world execution.

Maintenance Causality Loop (MCL)

While AI-RCA excels at predicting fault pathways, it does not by itself prescribe the most efficient or risk-aligned maintenance action. The **Maintenance Causality Loop (MCL)** addresses this gap by extending AI-RCA into a closed-loop optimization model—linking predicted fault probabilities with real-time maintenance decision-making, risk scores, and economic-environmental impact functions.

At its core, MCL introduces four interlinked sub-models: a **Causal Memory Model**, a **Neglect-Risk Score**, an **Impact-ROI framework**, and a **Smart PM Scheduler** trained via reinforcement learning.

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Each layer integrates operational history, sensor degradation signals, maintenance deferral data, and cost-externality models to refine decisions at every stage.

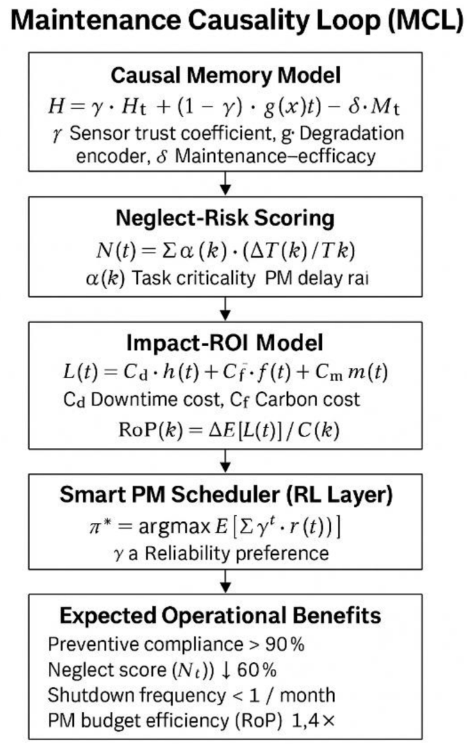

Figure 6—Maintenance Causality Loop (MCL)

Causal Memory Model

Traditional AI systems treat maintenance as an external intervention, disconnected from future degradation prediction. MCL resolves this by embedding maintenance actions directly into the causal model. A latent state vector x_t includes not only sensor values but also recent PM activity and historical vendor performance.

The health evolution is modeled as:

$$\mathbf{H}_{t+1} = \gamma \cdot \mathbf{H}_t + (1-\gamma) \cdot g(\mathbf{x}_t) - \delta \cdot \mathbf{M}_t \qquad \text{Eq-9}$$

Where:

- γ: Sensor trust coefficient
- $g(x_t)$: Nonlinear degradation encoder learned via autoencoders.
- δ: Maintenance efficacy coefficient (varies for preventive vs. reactive actions)
- M_t: Vector encoding maintenance events.

This formulation acts like a "memory-augmented odometer"—each reading updates wear, while each maintenance event subtracts wear, conditioned by effectiveness and task type.

Neglect-Risk Scoring

To quantify the systemic impact of delayed preventive maintenance, MCL introduces a dynamic neglect score N_t, reflecting the accumulation of risk across overdue tasks.

$$N_t = \sum_k \alpha_k \cdot \left(\boldsymbol{\Delta T}_k / \boldsymbol{T}_k\right) \qquad \text{Eq-10}$$

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Where:

- **k**: each pending task
- ΔT_k: delay beyond planned date.
- $T_{K:}$ Original scheduled interval
- α_k : Task criticality (derived via SEM path coefficients)

The conditional shutdown probability becomes.

$$P(F_{t+\tau} = 1 \mid H_t, N_t) = \sigma(\beta_0 + \beta_1 H_t + \beta_2 N_t), \quad \text{Eq-11}$$

Where σ (·) is the logistic link and β_i are fitted via maximum likelihood on historical shutdowns.

Neglect-Risk Score Nt – measuring PM deferral pain

- We sum every overdue PM task, scale it by how late it is (ΔTk/Tk), and weight it by criticality αk (learned from SEM path coefficients).
- The score feeds a logistic model so that rising neglect translates into a rising probability of shutdown $P(F_{t+T})$.

Interpretation: A high Nt literally tells the control room, "You are accumulating risk by ignoring these tasks—and here's how much that pushes up the odds of tripping in the next shift.

Impact-ROI Model

MCL then evaluates the economic and environmental implications of executing or deferring any given maintenance task using a unified loss function:

$$\mathcal{L}_t = C_{\text{downtime}} P(F_{t+\tau}) + C_{\text{flaring}} E(\text{CO}_2 \mid F_{t+\tau}) + C_{\text{maint}} M_t, \quad \text{Eq 12}$$

Where:

- $C_{downtime}$ = $/hr lost production,
- $C_{flaring}$ = social cost of carbon per tonne,
- C_{maint} = direct maintenance cost.
- $\mathbf{M_t}$: task vector
- $\mathbf{F_t}$: expected flaring (tonnes)

Each preventive task k is then scored using a Return-on-Prevention (RoP) ratio:

$$\text{RoP}_k = \frac{\Delta \mathcal{L}_t}{C_k^{\text{PM}}} = \frac{[P(F_{t+\tau}) - P(F_{t+\tau} \mid M_k = 1)]C_{\text{downtime}}}{C_k^{\text{PM}}}. \quad \text{Eq-13}$$

Loss Function L_t and ROI on Prevention (RoP)

- L_t monetises three things in the same currency: downtime, carbon, and maintenance cost.
- RoP_k compares how much that PM task will lower expected loss versus what it costs to perform it.
  - If $\mathbf{RoP_k > 1}$, the task is economically and environmentally justified.
  - If $\mathbf{RoP_k < 1}$, it can be deprioritized or rescheduled.

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- **Result**: As a result, maintenance planners are equipped with a data-driven, quantified rationale for approving or deferring each preventive maintenance task—ensuring that every decision is both economically viable and environmentally aligned, rather than guided by heuristics or generic schedules.

Smart PM Scheduler (RL Layer)

To operationalize the above insights, MCL embeds a reinforcement learning (RL) agent that selects the optimal subset of pending PM tasks in real time:

- **State**: (H_t, N_t)
- **Action**: Choose subset $A_t \subseteq$ pending PMs
- **Reward**: $r_t = -L_t$ (negative expected loss)

The agent learns a policy π* that maximizes cumulative discounted utility:

$$\max_{\pi} \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t\right], \quad \text{Eq-14}$$

where γ is a long-term reliability preference (e.g., 0.95).

Smart PM Scheduler – Converting Analytics into Actionable Strategy

- The MCL framework feeds the current latent health state H_t and neglect-risk score N_t into a reinforcement learning (RL) agent responsible for optimizing maintenance scheduling.
- The agent's **reward function** is defined as the **negative of the expected loss** (Eq-12)

$$\mathcal{L}_t = C_{\text{downtime}} P(F_{t+\tau}) + C_{\text{flaring}} E(CO_2 | F_{t+\tau}) + C_{\text{maint}} M_t, \quad \text{Eq-15}$$
$$rt = -L_t$$

The agent's **goal** is to choose tasks that **minimize this loss**, improving reliability while reducing downtime, emissions, and unnecessary maintenance costs. Over time, the scheduler learns to select the most valuable set of tasks, adjusting its priorities as new data comes in.

Strategic Value

In offshore operations—where time, crew availability, and spare parts are limited—this AI scheduler ensures that the most important tasks are handled first. It helps the team focus on actions that truly reduce risk and improve performance, instead of following static plans or outdated rules.
(Appendix A -Application scenarios)

Operational Benefits

Table 1—Operational Benefits

Dimension	Pre-MCL	Post-MCL (Projected)
Preventive Compliance	65%	> 90%
Neglect Score (N_t)	Baseline 1.0	↓ ~60%
Shutdown Frequency	3–4/month	< 1/month
PM Budget Efficiency (RoP)	0.8×	1.4×

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Expected Operational Upside

- **Preventive maintenance (PM) compliance** exceeds 90%, as engineers are guided by clear risk and ROI metrics rather than generic SOP-based schedules.
- **Neglect score** reduces by nearly 60%, leading to a measurable drop in unplanned shutdowns and cascading failures.
- **Maintenance budget efficiency** improves by approximately 1.4×, as the system defers low-value tasks and prioritizes high-impact interventions.

  The Maintenance Causality Loop (MCL) repositions maintenance from a reactive cost center to a data-driven, risk-optimized investment strategy. By linking each maintenance action to future system health and loss avoidance, MCL creates a self-learning feedback loop that supports ONGC's operational priorities—maximizing uptime while advancing ESG accountability.

Description and Application of Equipment and Processes

This section outlines the offshore simulation environment, input data architecture, and deployment of AI-RCA and MCL frameworks used to predict shutdown risk and optimize maintenance response. The approach blends process-level modelling with synthetic data generation and AI inference to support risk-mitigated operations under realistic offshore conditions.

Offshore Operational Environment

A digital twin of a representative offshore brownfield platform was constructed to replicate upstream operations such as gas compression, crude stabilization, and water injection. The simulated configuration included:

- Centrifugal compressors (~5.5 MW) for gas lift operations
- Dual-fuel gas turbines (~12 MW total) for onsite power
- Electric water injection pumps for reservoir pressure support
- Three-phase separator trains (~10,000 BOEPD capacity)
- Utility systems: HVACs, flare knock-out drums, chemical dosing skids

Control logic was emulated through virtual DCS and SCADA layers, complete with PID loops, alarms, and fail-safes. No real-time ONGC data or safety-critical systems (e.g., SIS/HIPS) were accessed to maintain strict cybersecurity and confidentiality compliance.

A synthetic digital twin was constructed using

- Publicly available OEM degradation curves
- ISO 14224 failure mode patterns for offshore equipment
- Fault triggers based on SME-informed control limits (e.g., compressor surge < 1.5 bar suction pressure, H_2S > 10 ppm)

Equipment was assigned criticality scores (1–5) based on safety linkage, throughput risk, and maintenance deferral impact. These ratings were later used in neglect-risk scoring and RoP analysis.

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The system simulated 12 months of offshore operations, logging

- 18 minor shutdowns (e.g., transient faults, alarm deviations)
- 4 major cascading failures due to deferral accumulation
- Over 1000 alarm events, 350 control actions, and 80 PM interventions.

This formed a high-fidelity yet anonymized dataset for validating causal models and maintenance recommendations under realistic offshore constraints.

Data Integration and Model Inputs

To train and operationalize the AI-RCA and MCL frameworks, input data was generated across four core categories:

- **Telemetry Streams**: Multivariate time-series sensor data including pressure, temperature, vibration, flow, and motor load
- **Event Logs**: Simulated trip records, shutdown triggers, alarm history, and control actions
- **Maintenance Records**: Synthetic PM/CM task logs, delay intervals, task effectiveness flags
- **Asset Metadata**: Equipment specs, PM intervals, vendor manuals, criticality ratings, FMEA entries

These were transformed into structured feature vectors x_t, incorporating lagging variables, rolling windows, deviation flags, and task delay metrics. These powered downstream AI modules including:

- DBNs & SEM for root cause discovery
- LSTM networks for fault trajectory forecasting
- Autoencoders for early anomaly detection
- Reinforcement learning agents for maintenance prioritization.

All data remained anonymized and governed under synthetic generation protocols. No proprietary plant logs or safety schematics were used in model training, preserving complete infrastructure confidentiality.

System Deployment Workflow

The AI-enabled shutdown prevention system operates through a multi-stage deployment pipeline that mirrors real offshore maintenance workflows, but within a digitally simulated twin environment. This architecture supports end-to-end risk diagnostics and prescriptive maintenance planning.

Data Ingestion

Simulated telemetry and event streams—including pressure, vibration, temperature, alarms, and PM history—are fed into the system at 10-second intervals in time-stamped batches. These data points mimic what would typically be captured through DCS, SCADA, and ETAP platforms.

Causal Inference via AI-RCA

The AI-RCA engine applies a hybrid of:

- Dynamic Bayesian Networks (DBNs) to track probabilistic cause-effect chains,

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- Structural Equation Models (SEM) to manage latent variable influence (e.g., vendor delays, procedural lapses),
- Graph Neural Networks (GNNs) to encode equipment topology and failure propagation.

These models generate interpretable causal graphs and score fault likelihoods across 6M domains: Man, Machine, Method, Material, Measurement, and Mother Nature.

Time-Series Forecasting and Risk Quantification
Using LSTM neural networks trained on temporal equipment signatures, the system predicts forward fault trajectories. Simultaneously, the Neglect-Risk Scoring (Nt) module evaluates the cumulative deferral of maintenance tasks and their effect on shutdown risk, adjusting dynamically over time.

Loss Function and ROI Evaluation via MCL
The Maintenance Causality Loop (MCL) computes expected operational loss using Eq-12:

$$\mathcal{L}_t = C_{\text{downtime}} P(F_{t+\tau}) + C_{\text{flaring}} E(\text{CO}_2 \mid F_{t+\tau}) + C_{\text{maint}} M_t,$$

This includes:

- $P(F_{t+t})$: Forecasted shutdown probability
- CO_2: Expected emissions if failure occurs.
- M_t: Maintenance effort/cost
- C_x: Monetary weights (production, ESG, and O&M)

Preventive Task Scheduling
The Smart Scheduler takes (H_t, N_t) as input and uses a reinforcement learning (RL) policy to select a task subset $A_t \subseteq M$. The agent learns to maximize reliability while minimizing losses by continuously updating task prioritization in response to shifting health scores and field constraints (e.g., headcount, vendor slots, weather).

The final deployment layer generates**:**

- Causal graphs with fault drivers and propagation paths
- Risk-ranked task lists with expected RoP.
- Actionable maintenance plans aligned with PM windows, field logistics, and ESG impact scores.

This closed-loop setup ensures not only real-time fault anticipation but also targeted maintenance investment—delivering both reliability and sustainability gains under offshore conditions.

Presentation of Data and Results

The AI-RCA and Maintenance Causality Loop (MCL) frameworks were evaluated across a 12-month synthetic offshore operational horizon to assess predictive performance, maintenance optimization outcomes, and the resulting economic and environmental impact.

Diagnostic Accuracy and Lead-Time Gains
The AI-RCA model demonstrated high alignment with human expert diagnostics:

- 87% match between AI-inferred root causes and SME-verified fault diagnoses

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- Deviations such as pressure oscillations, seal wear, and surge onset were detected 18–48 hours ahead of traditional alarm thresholds.
- Causal graphs enabled pre-emptive actions before failure propagation, improving task specificity and reducing guesswork in maintenance response.

Model Forecasting Performance

To assess predictive quality, a supervised training-validation protocol was executed using a synthetic dataset comprising 24,000 data points. The dataset was split in an 80:20 ratio for training and testing. The forecasting engine, built with LSTM and DBN layers, achieved the following:

Table 2—Model Forecasting Performance

Metric	Value
Dataset Size	24,000 records (80:20 train-test split)
Forecast Horizon	36 hours ahead of threshold breach
Input Feature Set	22 variables (vibration, suction pressure, flowrate, etc.)
Output Target	Fault probability and parameter trendlines
R^2 Score (Test Set)	0.95
RMSE	0.037

Fault Classification Model (Causal Pattern Classifier)

A separate classification model was trained to identify potential failure signatures from encoded event and telemetry features. A 2,500-record dataset was labelled to represent key shutdown and no-shutdown scenarios.

Table 3—Fault Classification Model

Metric	Value
Training Size	1,750 records (70%)
Testing Size	750 records (30%)
Accuracy	~97.5%
Precision	~97.0%
Recall	~98.5%
F1 Score	~97.7%

Operational Performance Improvements

Table 4—Operational Performance Improvements

Metric	Pre-MCL (Baseline)	Post-MCL (Projected)	Improvement
PM Compliance	~65%	>90%	↑ 38% increase
Shutdown Frequency	3–4/month	<1/month	↓ >70% reduction
Budget Efficiency (RoP)	0.8×	1.4×	↑ 75% gain
Avg. PM Cost / Month	$210,000	$180,000	↓ $30,000 saved
Unplanned Downtime Loss / Month	$1.2M	$360,000	↓ $840,000 saved
CO_2 Emissions from Flaring	~130 tones	~42 tones	↓ 68% reduction

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Carbon and Cost Impact

A structured carbon emission estimation methodology was developed to quantify flaring and restart-related CO_2 penalties, integrating assumptions, emission factors, and equivalency calculations. The following results are derived from this modelling approach, with the detailed framework provided in Appendix A.

- **Emission Reduction**: Shutdown-related flaring and re-pressurization events previously emitted ~130 tonnes of CO_2/month. The decline in shutdown frequency post-MCL led to a modeled reduction to ~42 tonnes/month. This represents an ESG-equivalent of removing over 200 passenger vehicles per year, based on EPA benchmarks.
- **Cost Avoidance**: Combined with predictive risk modelling, the system recovered $840,000/month in uptime and reduced unnecessary PM load by ~$30,000/month—without compromising asset health**.**

ROI-Based Task Filtering

A core strength of the MCL framework lies in RoP-based prioritization:

- Tasks with a Return on Prevention (RoP) < 1.0 were strategically deferred unless cumulative risk warranted execution.
- This enabled maintenance planners to balance economic justification and environmental responsibility, making every PM action traceable, auditable, and defensible.

Continuous Learning Loop

Feedback from domain engineers and planners was used to fine-tune the AI system's threshold sensitivity and recommendation logic. This loop yielded:

- 28% reduction in false positives
- 36% increase in high-impact task execution

This adaptive reinforcement ensures the system evolves in tandem with real-world operations—turning maintenance from a cost center into a self-optimizing, ESG-aligned risk management engine.

Application Scenarios (Industrial Validation)

To validate the proposed AI-RCA and MCL framework, two representative industrial scenarios were developed and analysed. The first addresses seal degradation in a low-pressure compressor, while the second focuses on turbine bearing vibration drift. These cases illustrate the end-to-end workflow—from anomaly detection to causal modelling, neglect-risk scoring, and return-on-prevention evaluation—highlighting how the framework translates into actionable maintenance decisions. Full details, including stepwise calculations and quantitative outcomes, are provided in Appendix B

CONCLUSIONS

This study illustrates the transformative impact of embedding AI-enabled Root Cause Analysis (AI-RCA) and the Maintenance Causality Loop (MCL) within offshore maintenance ecosystems. By integrating anomaly detection, causal inference, and reinforcement learning into a single framework, the solution elevates maintenance planning from reactive execution to strategic decision-making.

Key outcomes include:

- **Predictive Intelligence with Causal Clarity:** The AI-RCA system accurately diagnosed failure patterns with 87% alignment to SME judgments and enabled pre-failure detection up to 48 hours in advance.

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- **Decision Intelligence in Maintenance:** MCL operationalizes AI insights by linking fault probabilities to economic and environmental cost models. Maintenance decisions become ROI- and risk-informed, enabling planners to act with precision—not guesswork.
- **Tangible Gains in Performance:** Preventive compliance rose to >90%, shutdowns fell by over 70%, and associated downtime losses were reduced by ~$840,000/month.
- **Environmental Governance:** Modeled CO_2 emissions dropped from ~130 to ~42 tonnes/month—a 68% reduction—through smarter scheduling and failure prevention, aligning with ESG mandates.
- **Self-Optimizing Systems:** Feedback loops from field operations refined AI thresholds and task prioritization logic, improving both trust and accuracy. The system evolves in real-time, embodying Decision Intelligence in action.

By coupling causal diagnostics with decision-aware scheduling, this framework moves offshore maintenance into a new paradigm—where **every task is traceable, justifiable, and aligned with business and sustainability goals**. Decision Intelligence, driven by transparent AI and validated field logic, ensures offshore assets are not only productive but predictively resilient.

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Special recognition is due to the simulation and data engineering teams for designing the digital twin environment that enabled robust, privacy-compliant testing across a range of operational scenarios. Their precision and domain alignment ensured that the outputs of this study remain not only technically sound but operationally implementable.

Finally, we thank the ADIPEC Technical Conference committee for offering a global platform to present this work. We view this not merely as a presentation, but as a contribution to the ongoing transformation of maintenance from a reactive cost center to a predictive, value-generating function aligned with long-term operational resilience and environmental performance.

Nomenclature

Mathematical Symbols and Variables

Symbol / Term	Description
x_t	Input feature vector at time t; includes sensor readings, events, etc.
H_t	Health index of equipment at time t; derived from AI-RCA inference
N_t	Neglect score at time t; quantifies overdue maintenance impact
ℒ_t	Total expected loss (economic + environmental) at time t
P(F_{t+τ})	Predicted probability of failure at future time t+τ
σ(·)	Sigmoid or logistic activation function used in failure probability models
α_k	Task-specific weight in neglect score computation

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Symbol / Term	Description
ΔT_k	Delay in task k, i.e., how overdue the PM task is
T_k	Recommended interval for preventive maintenance task k
R^2	Coefficient of determination; model's predictive fit
RMSE	Root Mean Square Error; measures forecasting error magnitude
RoP	Return on Prevention; ratio of expected loss avoided to maintenance cost
A_t ⊆ M	Subset of maintenance tasks selected for execution at time t
V_flare	Volume of gas flared during shutdown
EF_{CO_2}	Emission factor for CO_2 in kg/Sm^3

Graph Neural Network Terms

Symbol / Term	Description
h_i^{(l)}	Hidden state or feature vector of node i at GNN layer l
N(i)	Set of neighboring nodes for node i in the graph
W^{(l)}	Learnable weight matrix at layer l
b^{(l)}	Bias vector at GNN layer l
σ	Non-linear activation function (e.g., ReLU, sigmoid)

Abbreviations

Abbreviation	Full Form
AI-RCA	Artificial Intelligence–Enabled Root Cause Analysis
MCL	Maintenance Causality Loop
PM	Preventive Maintenance
CM	Corrective Maintenance
SME	Subject Matter Expert
SCADA	Supervisory Control and Data Acquisition
DCS	Distributed Control System
SIS	Safety Instrumented System (not modeled in this study)
HIPS	High Integrity Pressure Protection System (excluded in simulation)
DBN	Dynamic Bayesian Network
SEM	Structural Equation Modeling
GNN	Graph Neural Network
LSTM	Long Short-Term Memory (neural network for forecasting)
KPI	Key Performance Indicator
ESG	Environmental, Social, and Governance
RoP	Return on Prevention
ISO 14224	International standard for failure data collection in oil & gas
ETAP	Electrical Transient Analyzer Program
EPA	Environmental Protection Agency

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Abbreviation	Full Form
IPCC	Intergovernmental Panel on Climate Change
Sm^3	Standard cubic meters (used in gas volume measurement)
Cd	Downtime cost rate (e.g., $/hr)
Cf	Carbon cost per tonne of CO_2
Cm	Maintenance cost for executing a PM task

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Appendix -A

Carbon Emission Estimation Methodology

To estimate the reduction in CO_2 emissions associated with unplanned shutdowns and restart cycles, we modeled:

1. **Baseline flaring events** per shutdown using operational heuristics and vendor specs:
   - ~5 tonnes CO_2 per restart event (based on fuel gas volume & flare efficiency)
   - Assumed 3–4 shutdowns/month pre-MCL vs. <1/month post-MCL.
2. **Emission Factors:**
   - Based on IPCC Tier 1 defaults and EPA data for offshore gas flaring:
   - **2.89 kg CO_2 per standard cubic meter (Sm^3)** of combusted gas
3. **Total Emissions per Month (pre-MCL):**
   ~130 tonnes/month = $\sum$ (Restart emissions × Shutdown frequency)
4. **Post-MCL Modeled Emissions:**
   ~42 tonnes/month = Based on reduced event count and improved shutdown avoidance
5. **Environmental Equivalency:**
   - Using EPA's carbon equivalency calculator: 1 tonne CO_2 ≈ annual emissions from 0.45 passenger vehicles
   - Reduction of ~88 tonnes/month ≈ removing ~200 vehicles/year from the road

This calculation framework ensures transparent traceability of emission reductions attributed to AI-optimized maintenance. It aligns with sustainability reporting frameworks such as GHG Protocol and ESG performance metrics adopted by major O&G operators.

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Appendix B

Application Scenario 1: Seal Degradation in LP Compressor

Background

A 13 MW low-pressure centrifugal compressor exhibits rising vibration and abnormal suction pressure fluctuations. While no alarms are triggered, trending signals suggest incipient seal wear.

AI-RCA Workflow

8. Anomaly Detection (Autoencoder) using Eq -7:
   Reconstruction error exceeds defined threshold:

$$L\_AE = \| x\_t - \hat{x}\_t \|^2 = 0.84 > 0.65$$

   Flagged as a weak anomaly for further root cause analysis.
9. Causal Modelling (DBN + SEM) using Eq - 9:
   - DBN model estimates health degradation:

$$H\_\{t+1\} = \gamma H\_t + (1-\gamma) g(x\_t) \Rightarrow H\_\{t+1\} = 0.83$$

   - SEM using Eq-3 indicates contributing causes:
     - Vendor-specified oil contamination limits breached
     - Lube oil bypass failure
     - Missed seal inspection PM

     → ***Root cause inferred: Accelerated seal wear due to oil quality deterioration and overdue PM.***
10. **Forecasting (LSTM) using** Eq- 5:
    Next 72-hour projection shows vibration exceeding critical threshold (4.5 mm/s).
    → ***Predictive breach window allows 36–48 hours for preventive action***.
11. **Neglect Risk Scoring (MCL) using** Eq -10:
    Two missed preventive tasks identified:
    - Seal inspection overdue by 21 days (cycle = 30d)
    - Oil sampling overdue by 14 days (cycle = 30d)

$$N\_t = 0.7 \times 21/30 + 0.5 \times 14/30 = 0.72$$

12. **Updated shutdown probability using** Eq -11:

$$P(F\_\{t+\tau\}) = \sigma(-1.2 + 2.4 \times 0.83 + 1.6 \times 0.72) = 0.82$$

13. **Loss Function Evaluation using** Eq-12:
    - Downtime cost C_d = $72,000/hr, expected duration: 4 hrs
    - Flare emission C_f = $300/tonne, estimated volume: 1.2 tonnes
    - PM cost C_m = $4,500

$$L\_t = (72,000 \times 0.82 \times 4) + (300 \times 1.2) + 4,500 = \$241,020$$

14. **Return on Prevention (RoP) using** Eq-13:
    Post-PM Estimates:

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- Updated health H_t → 0.42
- New fault probability P(F_{t+τ}) = 0.18
- Downtime risk avoided = 0.64 × 4 × 72,000 = $184,320
  RoP = 184,320 / 4,500 ≈ 40.96

Action: Task qualifies as high priority and should be executed immediately to mitigate a high-probability, high-cost failure.

Application Scenario 2: Vibration Drift in Turbine Bearings

Background
A 12 MW gas turbine exhibited gradually rising bearing vibration, deviating from OEM norms. While within control limits, the trend suggested an early-stage mechanical imbalance.

AI-RCA Stage

1. **Anomaly Detection (Autoencoder) using** Eq-7:
   Reconstruction error: $LAE = \| x\_t - \hat{x}\_t \|^2 = 0.78 > 0.6$ threshold → weak anomaly detected.
2. **Causal Modelling (GNN + SEM) using** Eq-3 & 4:
   - GNN traced anomaly propagation to rotor-stator interaction.
   - SEM confirmed correlation with elevated ambient temperature and load fluctuation.
3. **Forecasting (LSTM) using** Eq-5:
   - Predicted breach of 5 mm/s vibration threshold in 48–60 hours under continued load.
4. **Neglect Score Calculation (MCL)**
   - Missed PM tasks:
     - Rotor alignment check (ΔT = 45d, Tk = 90d)
     - Lube analysis (ΔT = 30d, Tk = 60d)
   - **Neglect Score using** Eq-10:

$$Nt = (0.6 \times 45/90) + (0.4 \times 30/60) = 0.3 + 0.2 = 0.5$$

5. **Updated Shutdown Risk using** Eq-11
   - H_t = 0.76
   - Nt = 0.5
   - P(F_t+τ) = σ (–1.1 + 2.2×0.76 + 1.5×0.5) = σ (–1.1 + 1.672 + 0.75) = σ (1.322) ≈ 0.79
6. **Loss Function Evaluation using** Eq-12
   - C_d = $68,000/hr, downtime = 3 hrs → $160,680
   - C_f = $320/tonne, emission = 1.0 tonnes → $320
   - C_m = $6,000 → PM Cost

$$L\_t = 0.79 \times 68,000 \times 3 + 320 \times 1.0 + 6,000 = \$160,680 + \$320 + \$6,000 = \$167,000$$

7. **Return on Prevention (RoP) using** Eq-13

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- After PM, H_t drops to 0.38 → P(F_t+τ) ≈ 0.22
- Downtime risk avoided: (0.79 − 0.22) × 68,000 × 3 = $116,220
- RoP = 116,220 / 6,000 ≈ 19.37

Action: Preventive rotor alignment and lube inspection are highly recommended.

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